Problem Set 1

Git, working with notebooks

This problem set contains three tasks to complete during class and two homework tasks marked 🏠:

1. Introduction to GitHub (2 pts)
2. Resolve merge conflicts (2 pts)
3. Working with notebooks (1 pt)
4. 🏠 Git basics (0 pts)
5. 🏠 Git (2 pts)

To get points for a task, show your solution to the instructor and answer a "check-up question". For this problem set, the instructor may ask, for example:

- How is a repository different from a regular folder?
- What is a commit, and why do we add a message to it?
- What is a branch, and when is it worth using one?
- When does a conflict occur? Does Git resolve conflicts on its own?
- How can you check whether a notebook is actually using the GPU?
- What happens in marimo when you change a value in one cell?

Task 1. Introduction to GitHub

Complete the tutorial: https://github.com/skills/introduction-to-github

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Task 2. Resolve merge conflicts

Complete the tutorial: https://github.com/skills/resolve-merge-conflicts

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Task 3. Working with notebooks

1. Download the Jupyter Notebook available on the course website.
2. Run it in Google Colab, using the GPU available there.
3. Convert the notebook to a `.py` file, and then to a marimo notebook.
4. Run the marimo notebook on molab.
5. If you are waiting for the instructor, browse one of the notebooks at https://molab.marimo.io/notebooks. Show the instructor something you find interesting :)

Task 4. 🏠 Learn Git Branching

If you have not used Git before, complete the interactive tutorial:
https://learngitbranching.js.org

Task 5. 🏠 Git

1. Complete tasks 1 and 2 using the `git` command in the terminal. Your solution does not have to use the minimum number of steps.
2. Upload your terminal command history (the `history` command) to the SKOS platform.